

MS:

(a)

$$f'(x) = 3x^2 - 1 = 0 \implies x^2 = \frac{1}{3} \implies x = \pm\sqrt{\frac{1}{3}}; \text{ only } \sqrt{\frac{1}{3}} \in [0, 1]$$

$$f''(x) = 6x \Big|_{x=\sqrt{\frac{1}{3}}} > 0 \implies \text{Minimum}$$

$$f(0) = 1, \quad f(1) = 1, \quad f(x^*) < 1$$

$x = 0, x = 1$ are global maxima and x^* is a global minimum.

(b)

$$f'(x) = 2x - 2 = 0 \implies x = 1$$

$$f''(x) = 2 > 0 \implies \text{Minimum}$$

$$f(0) = 17, \quad f(5) = 32, \quad f(x^*) = 16$$

Global Min at $x = 1$, Global Max at $x = 5$

(c)

$$f'(x) = \frac{2xe^x - x^2e^x}{e^{2x}} = \frac{x(2-x)}{e^x} = 0$$

$$\implies x = 0, x = 2$$

$$f''(x) = \frac{(2-2x)e^x - (2x-x^2)e^x}{e^{2x}} = \frac{x^2 - 4x + 2}{e^x}$$

$$f''(x) \Big|_{x=0} > 0 \implies \text{Min at } x = 0$$

$$f''(x) \Big|_{x=2} < 0 \implies \text{Max at } x = 2$$

Global Min at $x = 0$, Global Max at $x = 2$

(d)

$$f'(x) = -3 \implies \text{A line}$$

$$f(-3) = 11, \quad f(10) = -28$$

Global Max at $x = -3$, Global Min at $x = 10$

(e)

$$f'(x) = -2x + 6 = 0 \implies x = 3$$

$$f''(x) = -2 < 0 \implies \text{Max}$$

$$f(0) = 12, \quad f(10) = -28, \quad f(3) = 21$$

Global Max at $x = 3$, Global Min at $x = 10$

(f)

$$f'(x) = 9x^2 \geq 0 \implies x = 0$$

$$f''(x) = 18x$$

$$f'''(x) = 18 \implies \text{Inflection Point}$$

$$f(0) = -4, \quad f(10) \geq 0$$

Global Min at $x = 0$, Global Max at $x = 10$

(g)

$$f'(x) = 12x^2 + 2x - 2 = 0 \iff x^2 + \frac{1}{6}x - \frac{1}{6} = 0 \implies x = -\frac{1}{2}, x = \frac{1}{3}$$

$$f''(x) = 24x + 2 \implies f''(-\frac{1}{2}) < 0 \implies \text{local max}$$

$$f''(\frac{1}{3}) > 0 \implies \text{local min}$$

$$f(-1) = 2, \quad f(1) = 6, \quad f\left(-\frac{1}{2}\right) = \frac{15}{4}, \quad f\left(\frac{1}{3}\right) = \frac{70}{27}$$

Global Max at $x = 1$, Global Min at $x = -1$

(h) $x = -\frac{1}{2}$ is a max and $x = \frac{1}{3}$ is a min.

$$f\left(-\frac{1}{2}\right) = \frac{15}{4}, \quad f\left(\frac{1}{3}\right) = \frac{70}{27}, \quad f(-3) = -90, \quad f(3) = 114$$

Global max at $x = 3$ and global min at $x = -3$

Additional Questions:

1. A firm has two different plants. Let the cost at plant 1 be given by $c(y_1) = 5y_1^2 + 100y_1$, and the cost at plant 2 by $c(y_2) = 500y_2$. Suppose the firm wants to produce $\bar{Q}$ units of total output.

Cont.

(a) Total Cost= $TC = c(y_1) + c(y_2) = 5y_1^2 + 100y_1 + 500y_2$.

(b) Marginal cost at plant 1: $MC_1 = \frac{\Delta c(y_1)}{\Delta y_1} = 10y_1 + 100$

Marginal cost at plant 2: $MC_2 = \frac{\Delta c(y_2)}{\Delta y_2} = 500$

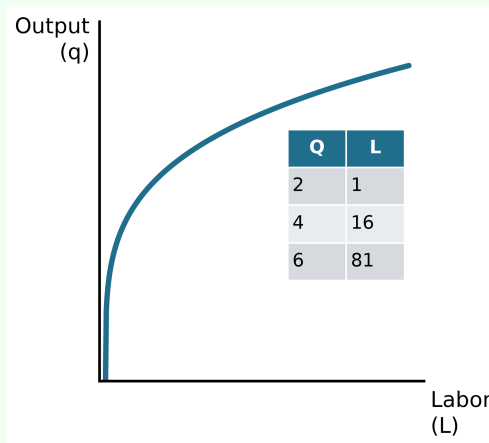
(c) The cost minimizing level of output should satisfy $MC_1 = MC_2$. If not, you should shift production to the least costly plant. Let $y_1 + y_2 = \bar{Q}$

$$\begin{aligned} \implies 10y_1 + 100 &= 500 \implies y_1 = \frac{400}{10} = 40 \\ y_2 &= \bar{Q} - 40 \quad (\text{assuming } \bar{Q} \geq 40; \text{ if } \bar{Q} < 40, \text{ then } MC_1 < 500 \text{ throughout, so} \\ &\text{plant 1 produces everything: } y_1 = \bar{Q}, y_2 = 0) \end{aligned}$$

(d) Plant 1 should increase production until $y_1 = 40$.

2. *Silky Smoothies* production process can be described in the following way $F(L, K) = L^{\frac{1}{4}} K^{\frac{1}{2}}$. Where L and K broadly define capital and labor inputs used in the production of smoothies. Suppose K is fixed at 4 units. Graph *Silky Smoothies* total product of labor curve. For what values of L does this firm exhibit diminishing returns to labor? In practice, why would a firm exhibit diminishing returns to labor?

When $K = 4 \implies F(L, 4) = 2L^{\frac{1}{4}}$



The firm examines diminishing returns, and each additional worker still increases output, but by a smaller amount than the previous one. To determine where the firm experiences diminishing marginal returns to labor, we compute the **marginal product of labor** (MP_L), which is the first derivative of output Q with respect to labor L :

$$MP_L = \frac{dQ}{dL} = \frac{d}{dL} \left(2L^{\frac{1}{4}} \right) = 2 \cdot \frac{1}{4} L^{-\frac{3}{4}} = \frac{1}{2} L^{-\frac{3}{4}}$$

To check for diminishing returns, examine whether the marginal product decreases as L increases $\implies$ we take the derivative of MP_L with respect to L :

$$\frac{dMP_L}{dL} = \frac{d}{dL} \left(\frac{1}{2} L^{-\frac{3}{4}} \right) = \frac{1}{2} \cdot \left(-\frac{3}{4} \right) L^{-\frac{7}{4}} = -\frac{3}{8} L^{-\frac{7}{4}}$$

Since $L^{-\frac{7}{4}} > 0$ for all $L > 0$, it follows that:

$$\frac{dMP_L}{dL} < 0 \quad \text{for all } L > 0$$

The firm exhibits *diminishing returns to labor* for all positive values of L .

In practice, capital is fixed in the short run (here $K = 4$): each additional worker shares the same equipment and space, so every added worker raises output by less than the one before.

3. A firm faces the following demand curve: $P = 120 - 0.02Q$; where Q is weekly level of production and P is price, measured in cents per unit. The firm's cost function is given by $C = Q + 25,000$. Assume that the firm maximizes its profits. What is the level of production, price, and total profit per week?

$$\text{Revenue} = P \times Q \implies (120 - 0.02Q) \times Q = 120Q - 0.02Q^2$$

$$\text{Marginal Revenue: } MR = 120 - .04Q$$

$$\text{Marginal Cost: } MC = 1$$

$$\text{Profit max } Q^* \text{ at } MR = MC \implies 120 - .04Q = 1 \implies Q = 2975$$

$$\implies P = 120 - 0.02(2975) = 60.5$$

$$\implies \pi = 60.5(2975) - 2975 - 25,000 = 152,012.5$$